Modelling Moon Phases · Staff Answers

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Arrival Answers

1. The Sun.
2. The Moon.
3. A person viewing from Earth.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Without tilting the orbit slightly, the observer's head blocks the torch at Position 5 and creates a lunar eclipse instead of a Full Moon.

Hinge: A. Between First Quarter (Day 7) and Full Moon (Day 14), the lit share continues growing through Waxing Gibbous.

Evidence Desk Cards

Position 1 (Day 0): Ball directly toward the torch: New Moon: 0% of the lit side is visible from Earth.

Position 2 & 3 (Days 4 & 7): Turning anticlockwise: Waxing Crescent and First Quarter: the visible lit share is growing.

Position 4 (Day 10): Three-quarters of the face is lit: Waxing Gibbous: more than half of the disc is lit and still growing toward Full.

Position 5 (Day 14): Ball held opposite the torch: Full Moon: 100% of the sunlit half faces the observer.

Positions 6, 7 & 8 (Days 18, 21 & 25): Returning to Pos 1: Waning phases: Waning Gibbous, Third Quarter, and Waning Crescent.

Torch is smaller than the observer's head and only 2 metres away: Model limitation: the sizes and distances of the Sun, Earth, and Moon are not to scale.

Check & Repair Answer

Half of the 28-day orbit takes about 14 days, so the lit part shrinks gradually through the three waning phases.

Exit Ticket Answers

1. Torch = Sun, Observer = Earth, Ball = Moon. 2. New Moon -> First Quarter -> Full Moon -> Third Quarter (with Waxing/Waning Crescent and Gibbous in between). 3. Strength: shows in 3D that half the Moon is always lit while our viewing angle changes. Limitation: sizes and distances are not to scale, and Earth's daily spin/orbit motion are simplified.